PRIOR: Deep Reinforced Adaptive Video Streaming with Attention-Based Throughput Prediction

Danfu Yuan[1], Yuanhong Zhang[1], Weizhan Zhang[1*], Xuncheng Liu[1], Haipeng Du[1,2], Qinghua Zheng[1]
[1]*School of Computer Science and Technology, MOEKLINNS Lab,* [2]*School of Continuing Education, Xi'an Jiaotong University*

ABSTRACT

Video service providers have deployed dynamic video bitrate adaptation services to fulfill user demands for higher video quality. However, fluctuations and instability of network conditions inhibit the performance promotion of adaptive bitrate (ABR) algorithms. Existing rule-based approaches fail to guarantee accurate throughput estimates, and learning-based algorithms are considerably sensitive to the variability of network. Therefore, how to gain effective and stable throughput estimates has become one of the critical challenges to further enhancing ABR methods. To eliminate this concern, we propose PRIOR, an ABR algorithm that fuses an effective throughput prediction module and a state-of-the-art multi-agent reinforcement learning method to provide a high quality of experience (QoE). PRIOR aims to maximize the QoE metric by straightforwardly utilizing accurate throughput estimates rather than past throughput measurements. Specifically, PRIOR employs a light-weighted prediction module with attention mechanism to obtain effective future throughput. Considering the excellent features introduced by the HTTP/3 protocol, we apply PRIOR to trace-driven simulations and real-world scenarios over HTTP/1.1 and HTTP/3. Trace-driven emulation illustrates that PRIOR outperforms existing ABR schemes over HTTP/1.1 and HTTP/3, and our prediction module can also reinforce the performance of other ABR algorithms. Extensive results on real-world evaluation demonstrate the superiority of PRIOR over existing state-of-the-art ABR schemes.

CCS CONCEPTS

• **Information systems** → *Multimedia streaming*.

KEYWORDS

Throughput Prediction, Bitrate Adaptation, Reinforcement Learning, HTTP/3 Protocol

ACM Reference Format:
Danfu Yuan, Yuanhong Zhang, Weizhan Zhang, Xuncheng Liu, Haipeng Du, Qinghua Zheng. 2022. PRIOR: Deep Reinforced Adaptive Video Streaming with Attention-Based Throughput Prediction. In *32nd edition of the Workshop on Network and Operating System Support for Digital Audio and Video (NOSSDAV '22), June 17, 2022, Athlone, Ireland.* ACM, New York, NY, USA, 7 pages. https://doi.org/10.1145/3534088.3534348

*Weizhan Zhang is the corresponding author, ***zhangwzh@xjtu.edu.cn***

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
NOSSDAV'22, June 17, 2022, Athlone, Ireland
© 2022 Association for Computing Machinery.
ACM ISBN 978-1-4503-9383-6/22/06…$15.00
https://doi.org/10.1145/3534088.3534348

1 INTRODUCTION

Recent years have seen a rapid increase in the volume of Internet video streaming and downloads, which will grow to more than 80% percent of all the traffic on consumer Internet by 2022 [14]. Video service providers have struggled with optimizing the desired QoE objective by adjusting the parameters of video transmission. Rule-based ABR schemes, e.g., BB [8] and RobustMPC [27], fail to guarantee accurate throughput estimates while encountering various network conditions. When the estimation is inaccurate, especially if the requested quality exceeds the tolerance of the buffer under terrible network conditions, the future rebuffering can harm the QoE. Learning-based algorithms (e.g., Pensieve [11], Hot-DASH [22], and Stick [7]) achieve the state-of-the-art performances by utilizing the Deep Reinforcement Learning (DRL) framework to select the optimal bitrate for future video chunks. However, they are rather sensitive to dynamic bandwidth, resulting in an unsatisfactory real-world performance.

In light of these concerns, we observe a promising way to combine the efficiency of the DRL framework with synthesized network characteristics: we can leverage accurate throughput estimates to make learning-based methods more stable, and in turn, adopt the multi-agent DRL framework to output the optimal bitrates. Hence, this paper is motivated by a simple quest: can we utilize effective prediction methods to obtain the next throughput trajectory, aiming to make the learning-based scheme perform well under all the considered network conditions, and especially over HTTP/3. Unfortunately, despite the abundance of recently proposed approaches [23, 24], recent ABR methods with throughput prediction can hardly tackle all of the above challenges due to the inability to effectively utilize the throughput characteristics in the real-world network environment.

Further, many new features introduced by QUIC, such as multiplexing over a transport layer connection and lower latency handshake, make QUIC perform better than TCP in most scenarios, e.g., video transmission. HTTP/3, previously known as HTTP-over-QUIC [10], relies on a low-level UDP protocol in terms of reduced connection delay and features (e.g., congestion control, packet retransmission) inherited from the TCP protocol. Especially, considering the 0-RTT merit of QUIC, the client avoids waiting for a round trip time for each request before it can receive video segments, which means QUIC can improve the performance of video transmission. However, previous work [1, 2, 16] has not focused on deploying learning-based ABR algorithms over HTTP/3.

In this paper, we propose PRIOR, a novel ABR algorithm that integrates an effective throughput prediction module and the state-of-the-art multi-agent DRL to maximize the QoE. Aiming to accurately estimate throughput information, we further present GCA using **G**ated Recurrent Unit (GRU) and **C**onvolutional neural network (CNN) with **A**ttention mechanism. Specifically, GCA takes

historical observed throughput as the input, and utilize attention mechanism combining GRU and CNN to extract throughput characteristics. Unlike traditional learning-based approaches, PRIOR passes the synthesized throughput information from GCA module into the state's input, rather than a throughput sequence observed in the past several time stamps. Meanwhile, we adopt *proximal policy optimization* (PPO) with multi-agent, a discrete DRL method, to train the model from scratch. It follows the integral structure of PPO with single agent by learning a policy and a value function for each agent. Besides, it adopts the *centralized training and decentralized execution* (CTDE) structure. We implement an accurate value function to further improve policy learning through variance reduction by observing the full global state.

We design trace-driven simulations and real-world experiments over HTTP/1.1 and HTTP/3 to validate PRIOR's performance. First, we compare the performance of GCA module to off-the-shelf prediction approaches. The obtained results indicate that GCA maintains the best performance on both long-term and short-term predictions. Next, the comparison of PRIOR and existing ABR schemes shows that PRIOR surpasses recent work by 9.01% over HTTP/1.1 and 6.53% over HTTP/3 on average QoE. Moreover, we also prove that GCA can improve the performance of existing ABR schemes by improving the average QoE by 2.88% to 7.41%. Finally, we implement PRIOR and evaluate its performances under real-world environments, as expected, yielding reliable and solid results. To sum up, we summarize the contributions as follows:

(1) We propose GCA to answer how to use an attention-based model combining GRU and CNN to accomplish an accurate throughput prediction. Results indicate that GCA module can significantly improve the overall QoE of existing ABR schemes.
(2) We propose PRIOR, an ABR scheme which is the fusion of the DRL framework and effective throughput prediction approach. Trace-driven emulation indicates that PRIOR can achieve higher performances with higher bandwidth utilization compared to existing ABR algorithms.
(3) To the best of our knowledge, we are the first to apply learning-based ABR algorithms to the real-world environment over HTTP/3 protocol. Further, we prove the effectiveness of PRIOR under real-world environments.

2 RELATED WORK

Adaptive Video Streaming Algorithms. Recent ABR schemes are mainly composed of rule-based, learning-based and a combination of the two methods. BB [8] proposes a linear criterion threshold to adjust available buffer size. FESTIVE [9] predicts future throughput via the harmonic mean of the throughput measured for the past five chunk downloads. RobustMPC [27] employs Mixed Integer Linear Programming to calculate optimal bitrate based on buffer occupancy and network throughput. However, rule-based approaches are quite sensitive to tackle different network conditions. Pensieve [11] utilizes the A3C algorithm to select bitrates for future video chunks based on observations. Stick [7] harmoniously fuses the buffer-based and learning-based approach to reduce the computational costs while preserving the performances. Compared to the previous approaches, PRIOR not only takes full advantage of the available bandwidth, but also overcomes the highly variable

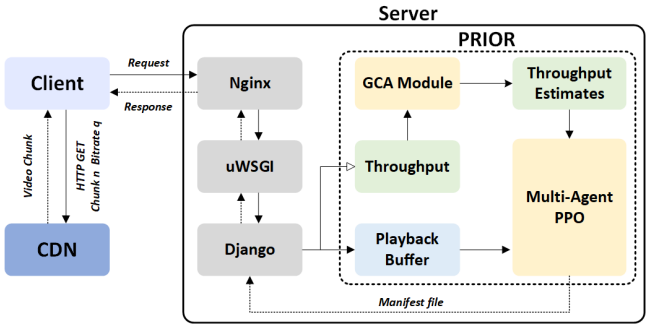

Figure 1: An overview of PRIOR's System Workflow.

throughput to avoid the degradation of QoE caused by rebuffering.

Throughput Prediction Approaches. Accurate throughput prediction is crucial for ABR approaches. The main throughput prediction methods can be divided into statistic-based and learning-based. Wei B et al. [26] establishes a parameterized autoregressive linear function to predict throughput. Zhang et al. [29] applies the Kalman filter approach to optimize throughput prediction. Besides, some researchers attempted to utilize machine learning methods to improve the performance of throughput prediction. LinkForeCast [28] develops a machine learning framework that utilizes both past throughput and lower-layer information to predict link bandwidth in real time. However, these approaches are rarely integrated with ABR algorithms, These methods are rarely combined with ABR algorithm, resulting in the inability to obtain higher QoE metric.

ABR over Different HTTP Protocols. The evolution of HTTP protocols and corresponding new features promote the optimization of ABR algorithm performance. Nikravesh et al. [15] leverages new features in HTTP/2 including stream multiplexing, flow control, and application-layer PING to boost the performance of QoE. Erfanian et al. [5] uses an SDN-based approach to maximize QoE objective and QoE-fairness in AVC-based HTTP adaptive streaming. Bhat et al. [1] finds that QUIC retransmissions can significantly improve the average quality bitrate while simultaneously minimizing bit rate variations over the duration of a streaming session. As we know, previous work has not focused on deploying learning-based ABR algorithms over HTTP/3 in the real-world environment.

3 PROPOSED METHODS

In this section, we describe the design and implementation of GCA and PRIOR. We start by proposing GCA module and explain the training methodology and algorithms underlying PRIOR. As shown in Figure 2, PRIOR is mainly composed of i) GCA module, which leverages attention mechanism to generate the next network trace trajectory, and ii) basic PRIOR, which takes effective throughput estimates as the input of neural network (NN) rather than the observed throughput, and adopts multi-agent reinforcement learning to optimize for different network characteristics and QoE objectives.

3.1 Motivation

Network conditions can vary significantly across environments and can fluctuate over time. This complicates bitrate adaptation as inaccurate throughput prediction may result in poor ABR performance.

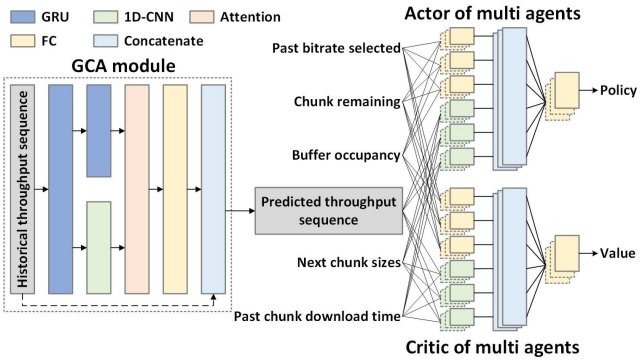

Figure 2: PRIOR's NN Architecture Overview.

For example, in time-varying network connections, throughput prediction by calculating the harmonic mean of past throughput measurements [9] is usually inaccurate and cannot account for sudden fluctuations in network bandwidth. Inaccurate predictions can result in network underutilization (decreased video quality) or increased download latency (rejection). Considering the violent fluctuations and periodic characteristics of the network throughput, rule-based ABR algorithms, e.g. RobustMPC's estimates are overly cautious and inaccurate, which prevent from reaching high bitrate even though the buffer occupancy continually increases [11]. Learning-based approaches [20] can hardly capture the hidden periodic and fluctuant characteristics in the throughput sequence and take the throughput measurements as NN inputs for training. Therefore, it has become a key challenge to obtain accurate throughput prediction in various real-world network conditions and help to improve the QoE performance in video adaptive transmission.

3.2 GCA Design

In this part, we propose a deep learning framework using GRU and CNN for throughput prediction based on attention mechanism.

Design. The network throughput is essentially a time series, so the prediction of throughput actually turns into a time series prediction problem. The NN-based methods have a great advantage in the prediction of time series problem [12]. It can make full use of the key information contained in the sequence to forecasts future network performance. Thus, we design a light-weighted NN composed of GRU and CNN to better capture periodic features and fluctuations hidden in historical throughput measurements. GCA's design are described in Figure 2.

Compared with Long Short Term Memory (LSTM), GRU can achieve considerable effect and greatly improve the training efficiency [3]. Moreover, CNN is used to extract the features of the throughput measurements. Due to the influence of various characteristics of historical throughput on the prediction accuracy, we introduce the attention mechanism to automatically capture the influence of feature states on the observed throughput at different times in the past to improve the prediction accuracy. The attention mechanism can be formulated as follows.

$$\alpha_i = softmax(V_s tanh(W_{ls} H'_{ti} + W_{cs} H^C_{ti})) \tag{1}$$

$$V_{attn} = \sum_{i=1}^{\omega} \alpha_i H^C_{ti} \tag{2}$$

where V_s, W_{ls}, W_{cs} are the coefficients of attention model respectively, α_i is the attention distribution weight, H'_{ti} and H^C_{ti} represent query vector and input vector generated by GRU and 1D-CNN, respectively. V_{attn} is the attention value obtained by weighted summation. GCA's workflow consists of three parts:

- Calculate the similarity or correlation between the query vector and input vector generated by GRU and CNN.
- Convert the score of the first stage numerically to highlight the weight of important elements through the internal mechanism of *softmax*.
- Calculate the attention value by weighted summation.

Implementation. GCA takes the past sequence length $k = 10$ into the NN. These sequential inputs are fed to a GRU with hidden size of 32. We use the same GRU and a 1D-CNN with 32 filters and the kernel size of 3. Results from GRU and 1D-CNN are aggregated in the attention layer. Next, we concatenate $h = 4$ past sequence and the outputs from a FC layer with 32 neurons. We set the predicted length p to 2, 4, 6, i.e., using the past $k = 10$ observed throughput to forecast future throughput of the next p seconds.

3.3 PRIOR Mechanism

This subsection describes PRIOR's design (Figure 2), structured according to how it addresses the two aforementioned challenges: how to model the state, action and reward? how to train PRIOR?

Design. The reinforcement learning model contains two interactive objects named agent and environment, and the interaction can be regard as a Markov decision process (MDP). In the process, it involves three elements, including the environment state, the action and its instant reward, represented as a tuple ⟨*state, action, reward*⟩. Through the constant interaction and iteration among them, the model finally learns an optimal bitrate adaptation strategy.

The state space includes observation from the environment related to the bitrate adaptation decision. Specifically, We first define the state space. In this paper, the state space includes the network throughput estimation T_p from GCA, the past bitrate selected q, the remaining number of chunks n, the current buffer occupancy b, the video sizes of the next chunk S, and the download time of the past chunks d. Here, the state s_k after downloading the k-th chunk can be define as $s_k = (T_p, q, n, b, S, d)$. PRIOR's learning agent takes the state s_k into NN. For the action space, it refers to the available bitrate for the next video chunk. The action of selecting the bitrate of the k-th chunk is expressed as a_k.

While the agent takes an action on the environment, the environment gives a reward as the feedback to indicate a value of the current action. And the state converts to a new state. We use the typical linear-based QoE metric used by MPC [27], and the QoE reward of the k-th chunk is formulated as follows.

$$QoE = \sum_{k=1}^{K} q(R_k) - \mu \sum_{k=1}^{K} T_k - \tau \sum_{k=1}^{K} |q(R_{k+1} - R_k)| \tag{3}$$

where K is the total chunks, R_k and T_k represent the video bitrate and the rebuffering time of the k-th chunk, $q(R_k)$ is a function that maps the bitrate R_k to the vidio quality perceived by the user. μ and τ are the penalty factors which control the priority of the given metric. Followed by recent work [11, 27], we set $q(R_k) = R_k$, $\mu = 4.3$ and $\tau = 1.0$ as the QoE baseline.

Training Methodology. The DRL-based ABR framework with GCA approach is shown in Figure 2. We exploit the state-of-the-art Actor-Critic RL algorithm PPO [21] with multi-agent to train the NN. PPO with multi-agent follows the algorithmic structure of PPO algorithm with single-agent by learning a policy π_θ and a value function $V_\phi(s)$ for each agent. In this paper, we utilize $V_\phi(s)$ in training for variance reduction to obtain global information, which can allow PPO with multi-agent to follow the CTDE structure. We design that all the agents share the actor and critic networks for great convergence. Each agent can also have its own pair of actor and critic networks, and follow common practices in PPO implementations, including gradient clipping, value clipping, ReLU activation, observation normalization and *Generalized Advantage Estimation* (GAE) with advantage normalization. Besides, to balance the exploration and exploitation of each agent, we set a sampling rule in Equation 4, where a_k is chosen to suit the environment.

$$a_k = \begin{cases} categorical_sample(\pi_\theta), & r > r_0 \\ argmax(cumsum(\pi_\theta)), & r \leq r_0 \end{cases} \tag{4}$$

where r is the random number in $(0, 1)$ and $r_0 = 0.5$.

The objectives of actor network and critic network are formulated as Equations 6 and 7, which are used to update the NN.

$$r_{\theta,i}^{(k)} = \frac{\pi_{\theta_{new}}\left(a_i^{(k)}|s_i^{(k)}\right)}{\pi_{\theta_{old}}\left(a_i^{(k)}|s_i^{(k)}\right)} \tag{5}$$

$$\begin{aligned} L(\theta) =& \frac{1}{MN}\sum_{i=1}^{M}\sum_{k=1}^{N}\min\left(r_{\theta,i}^{(k)}A_i^{(k)}, clip(r_{\theta,i}^{(k)})A_i^{(k)}\right) \\ &+ \sigma\frac{1}{MN}\sum_{i=1}^{M}\sum_{k=1}^{N}S(\pi_\theta(s_i^{(k)})) \end{aligned} \tag{6}$$

$$\begin{aligned} L(\phi) =& \frac{1}{MN}\sum_{i=1}^{M}\sum_{k=1}^{N}\max\left[\left(V_\phi(s_i^{(k)}) - \hat{R}_i\right)^2, \right. \\ &\left. \left(clip(V_\phi(s_i^{(k)}) - \hat{R}_i)\right)^2\right] \end{aligned} \tag{7}$$

where M is the batch size and N is the number of agents, $r_{\theta,i}^{(k)}$ is the ratio of the probability under the new and old policies, $A_i^{(k)}$ is the estimated advantage computed by the GAE method, S refers to the policy entropy, and σ refers to the entropy coefficient hyperparameter. The overall training precedure is described in Alg. 1.

Implementation. We leverage Tensorflow to construct the training process and implement the NN architecture. Specifically, we set sequence length of the input $k = 8$. For each feature in the input state, it passes through a 1D-CNN model with 128 filters and the kernel size of 4. Results from these layers are then aggregated with other inputs in a FC layer that uses 128 neurons. The final output of actor network is a probability calculated by the softmax function. The critic network uses the same structure and uses a linear neuron in the last layer.

4 EVALUATION

In this section, we prove the effectiveness of GCA module and evaluate PRIOR under various network conditions.

Algorithm 1 PRIOR Overall Training Procedure

```
Set learning rate α, batch size b_size
Initialize θ and φ, the parameters for policy π and critic V
while step < step_max do
    Set buffer B = {}
    for m = 1 to b_size do
        Set τ = {}
        Initialize actor states and critic states
        for t = 1 to T do
            for all agents a do
                p_t = π(o_t; θ)
                a_t ~ p_t
                v_t = V(s_t; φ)
            end for
            Execute actions a_t, observe s_{t+1}, o_{t+1}, r_t
            τ += [s_t, o_t, a_t, r_t, s_{t+1}, o_{t+1}]
        end for
        Compute advantage estimate Â via GAE on τ
        Compute reward-to-go R̂ on τ
        Divide trajectory τ into segments of length L
        for l = 0, 1, ..., T//L do
            D = D ∪ (τ[l : l + T, Â[l : l + L], R̂[l : l + L])
        end for
    end for
    for mini-batch k = 1, ..., K do
        b ← random mini-batch from D with all agent data
        for each data segment d in the mini-batch b do
            Update network for π and V
        end for
    end for
    Adam update θ on L(θ) with data b
    Adam update φ on L(φ) with data b
end while
```

4.1 Implementation

4.1.1 Experimental Testbed Setup. To better validate ABR schemes, we implement two experiments: trace-driven offline simulation and real-world experiment.

Environment. A server is committed to encode the video, store the video chunks and respond to the requests from clients. Our evaluations use the "Envivio-Dash3" video [6]. This video is encoded using the FFMPEG with H.264/MPEG-4 codec in this paper, and are available at 6 bitrates in $\{300, 750, 1200, 1850, 2850, 4300\}kbps$, corresponding to the resolutions $\{240, 360, 480, 720, 1080, 1440\}p$. The video are segmented into 4-second chunks, and the URLs are recorded in the MPD file.

Trace-driven Offline Simulation. We apply Mahimahi network emulation tool [13] to emulate different network conditions, especially setting mm-delay 40ms during the simulation along with different network traces. In our setup, PRIOR is integrated with the simulation environment.

Real-world Experiment. To evaluate the performance of the ABR schemes, we establish a system consisted of an ABR server, a video player and an HTTP content server. On the browser side, we modify dash.js to implement our video player. On the server side, PRIOR runs on a standalone ABR server, implemented using Django over uWSGI and Nginx-QUIC. The browser contacts the CDN to fetch the corresponding video chunk.

4.1.2 Datasets. To prove the effectiveness of GCA module, we create a corpus of throughput traces by collecting network traces from four public datasets: Ghent [25]: a 4G network dataset (collects 40 logs and covers 5 hours of active monitoring); UCC [18]: a 4G trace dataset (contains 135 traces, with an average duration of fifteen minutes per trace); FCC [4]: a broadband dataset (contains over 1 million throughput traces at a 5 second granularity); HSDPA [19]: a well-known 3G/HSDPA network trace dataset (contains comprises 30 minutes of throughput measurements). We generate 1000 traces for our corpus from the above datasets respectively by using a sliding window across the above datasets, each with a duration of 320 seconds. To avoid situations where bitrate selection is trivial, we only consider original traces with average throughput from 0.1Mbps to 8Mbps. We use a random sample of 70% of our corpus as a training set and the remaining 30% as a test set. Further, to evaluate PRIOR and state-of-the-art ABR algorithms on various network conditions, we generate 1000 traces from FCC [4] and HSDPA [19] datasets respectively, and each trace has a duration of 320 seconds. In the trace-driven emulations, we use 70% of traces for model training, 10% for validation and 20% for testing.

4.1.3 ABR Baselines. To examine the performance of PRIOR, we compare PRIOR to the following algorithms which collectively represent the state-of-the-art in bitrate adaptation.

(1) Buffer-Based (BB) [8]: dynamically selects bitrates with the goal of keeping the buffer occupancy above 5 seconds, and automatically chooses the highest available bitrate if the buffer occupancy exceeds 15 seconds. In this paper, we set the buffer bound $B = 5, 10$ as suggested by the authors.
(2) Rate-Based (RB) [9]: predicts throughput using the harmonic mean of the experienced throughput for the past 5 chunk downloads. It then selects the highest bitrate but lower than the predicted throughput.
(3) RobustMPC [27]: leverages control theory to optimize the QoE metric over a horizon of 5 future chunks by jointly considering the buffer occupancy observations and throughput predictions. We implement RobustMPC by ourselves.
(4) Pensieve [11]: firstly introduces RL-based techniques into adaptive video streaming systems, which employs the A3C architecture to select the optimal bitrate for the next video chunk. It learns to make ABR decisions solely through observations of the actual performance of past decisions. We use the pre-trained Pensieve model provided by the authors.
(5) Stick [7]: utilizes DRL to dynamically adjust the buffer-bound, and controls the ABR algorithm w.r.t. the current buffer-bound. It fuses the buffer-based method and learning-based method harmoniously. We set $B_{max} = 60$ as suggested by [11, 17] and implement Stick by ourselves.

4.2 Performance of GCA

To better understand how well GCA performs, we compare GCA with four representative algorithms, i.e., linear regression (LR), LSTM [12], bidirectional LSTM with attention (BiLSTM), and LinkForeCast [28] (LFCast). We set the prediction windows size k to 10 and the predicted length p to 2, 4, 6, i.e., using historical throughput in the past 10 seconds to predict the throughput of the next 2, 4 or 6 seconds. Figure 3 illustrates the *Root Mean Square Error* (RMSE)

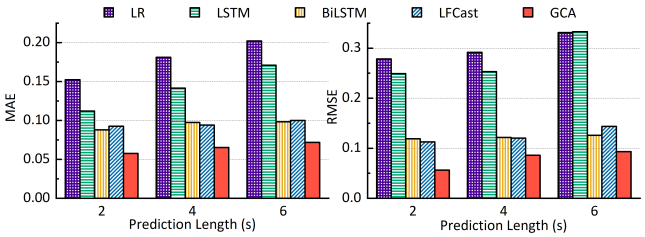

Figure 3: Comparing GCA with existing throughput prediction approaches.

and *Mean Absolute Error* (MAE) of the prediction results for each algorithm. We can find that when $k = 2$, GCA outperforms existing prediction approaches, with the average reduction on MAE of 32.21% to 63.55%. Besides, GCA betters on RMSE compared with existing algorithms, decreasing from 37.93% to 74.05%. Further, results also illustrate that using a larger predicted length p leads to a higher prediction error, and GCA still maintains the best performance on long-term predictions. Especially, GCA even works better than LFCast, by 28.26% to 30.61% on MAE and RMSE. The reason is that GCA considers the temporal and periodic features of network traces. Considering the trade-off between prediction accuracy and computation overhead, we use $p = 4$ in the rest of the paper.

4.3 PRIOR Evaluation

4.3.1 PRIOR vs. Existing ABR algorithms. In this part, we use trace-driven offline emulation over HTTP/1.1 to compare the performance of PRIOR against several existing ABR algorithms, including model-based ABR schemes (i.e., Buffer-based, Rate-based, RobustMPC) and learning-based ABR scheme (i.e., Pensieve, Stick), under FCC and HSDPA network dataset.

Figure 4 illustrates the CDF (Cumulative Distribution Function) distribution of average QoE performances for each ABR scheme. As expected, PRIOR outperforms existing algorithms, with the improvement on average QoE of 9.09% to 27.58%. Moreover, we also report the bitrate utility, rebuffering penalty as well as smoothness penalty, corresponding to the three parts of Equation 3. The obtained results indicate that compared PRIOR to other schemes, PRIOR always achieve higher average bitrate with lowest rebuffering. Note that PRIOR fails to obtain the highest bitrate and lowest bitrate changes, but results the highest average QoE. That means, PRIOR has generalized a strategy to build up sufficient buffer to handle the network's throughput fluctuations rather than blindly increase one of the underlying metrics. Same conclusion are also demonstrated in Pensieve [11].

4.3.2 Enhancement of GCA to ABR schemes. We set an experiment to better understand whether GCA help traditional ABR schemes improve the QoE performance, we validate three representative algorithms, i.e., RobustMPC, Pensieve and Stick. Table 1 illustrates, we observe that GCA improves the overall QoE of traditional heuristic ABR approaches, i.e., RobustMPC with the improvements on average QoE of 5.21%. Besides, GCA+Pensieve and GCA+Stick works better than the original schemes, by 2.88% to 7.41% on average QoE metric. The reason is that GCA module obtains effective future network throughput and helps model-based and learning-based ABR algorithms improve the bandwidth utilization up to 5.41%.

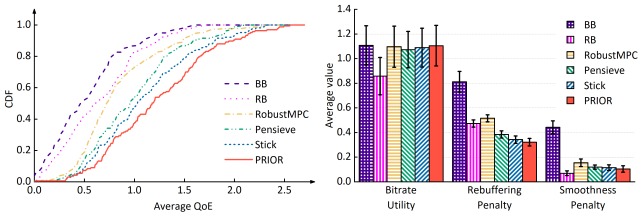

Figure 4: Comparing PRIOR with existing ABR approaches over HTTP/1.1.

Table 1: Enhancement of GCA to existing ABR schemes

ABR Schemes	Normalized Average QoE	Bandwidth Utilization (%)
RobustMPC	0.801	74.96
Pensieve	0.869	74.39
Stick	0.891	77.12
RobustMPC with GCA	0.845	79.24
Pensieve with GCA	0.894	77.83
Stick with GCA	0.957	80.97
PRIOR	1	83.02

4.3.3 ABR performances over different HTTPs. In this part, we investigate how PRIOR performs over different network access. We evaluate PRIOR and other ABR schemes over HTTP/3. Figure 5 illustrates the CDF of average QoE performances for each ABR algorithms, and we find that PRIOR maintains stable performance over HTTP/3, with the average QoE increasement of 6.53% and 4.01% compared to Pensieve and Stick. Comparing the metrics with Pensieve, PRIOR improves the average bitrate by 9.81%, decreases the average rebuffering time by 5.77%, and slightly increases average bitrate change by 4.03%. That means Prior attends to sacrifice smoothness to reduce rebuffering. Additionally, compared the QoE metrics over HTTP/1.1, each ABR scheme over HTTP/3 achieves higher average bitrate with lower rebuffering and bitrate changes. In particular, we observe that existing ABR schemes even improves the overall QoE of traditional heuristic approaches, i.e., buffer-based, rate-based and RobustMPC, with the improvements on average QoE of 7.02% to 9.84%. Besides, Pensieve, Stick and PRIOR improve 5.02%, 5.75% and 6.97%, respectively, in terms of average QoE from HTTP/1.1 to HTTP/3. That means the excellent characteristic of HTTP/3 protocol facilitates the QoE improvement.

4.4 PRIOR in the Real-world

In this experiment, we evaluate PRIOR as well as Pensieve under several types of network connections, and the detail of each link is listed as follows.

(1) Scenario A: a cross-city network route with average $RTT \approx 90ms$ (HTTP/1.1), $RTT \approx 60ms$ (HTTP/3), $packetloss < 3\%$. We fix the server's bandwidth to 2 Mbps.
(2) Scenario B: an international network link with average $RTT \approx 50ms$ (HTTP/1.1), $RTT \approx 36ms$ (HTTP/3), $packetloss < 5\%$.

Each link takes about 3 hours for each round. Results are reported with the average QoE of each session. As illustrated in Figure 5, PRIOR surpasses Pensieve over HTTP/1.1 and HTTP/3 in real world,

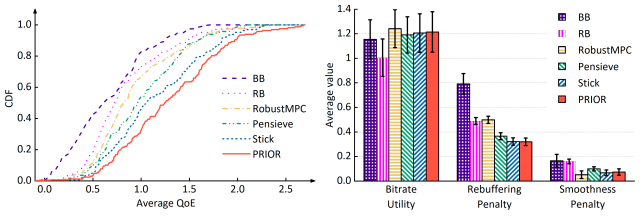

Figure 5: Comparing PRIOR with existing ABR approaches over HTTP/3.

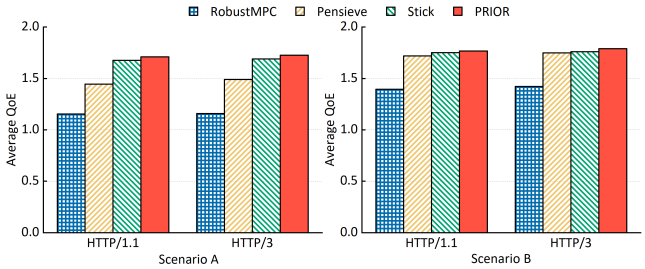

Figure 6: Comparing the average QoE of PRIOR with Pensieve and Stick under real-world network conditions.

with the improvements of 16.3% and 12.85% on average QoE, respectively. Such positive results indicates that PRIOR leverages an efficient throughput prediction module GCA to handle different network situations and applies these accurate throughput features to the multi-agent PPO algorithm. Moreover, ABR schemes over HTTP/3 performs comparable or better than these over HTTP/1.1. Through the experiment, we also confirm that PRIOR performs well even on these unstable network conditions.

5 CONCLUSION

In this paper, we present PRIOR, a novel ABR scheme fusing an efficient throughput prediction module and multi-agent reinforcement learning. First, we propose a tiny model named GCA to obtain the accurate future network throughput. Experimental results illustrates that GCA betters existing prediction algorithms on MAE and RMSE. At the same time, GCA also helps traditional heuristic ABR schemes and learning-based schemes improve the overall QoE up to 7.41%. Comparing to other traditional ABR algorithms, PRIOR faithfully utilizes the synthesized throughput information generated by GCA module as the input of the subsequent ABR controller. To better evaluate the efficiency of PRIOR, we apply PRIOR to the trace-driven emulation and real-world environment over HTTP/1.1 and HTTP/3. Results show that PRIOR outperforms existing ABR schemes on QoE metric, with increasement of 9.09% and 27.58% over HTTP/1.1. Meanwhile, existing ABR schemes improves 5.02% to 9.84% in terms of average QoE from HTTP/1.1 to HTTP/3.

ACKNOWLEDGMENTS

We thank the MMSys and NOSSDAV reviewer for the valuable feedback. This work was supported by National Key R&D Program of China (No. 2020AAA0108800), NSFC under Grant 62172326 and 62137002, the MOE Innovation Research Team No. IRT 17R86, and the Project of China Knowledge Centre for Engineering Science and Technology.

REFERENCES

[1] Divyashri Bhat, Rajvardhan Deshmukh, and Michael Zink. 2018. Improving qoe of abr streaming sessions through quic retransmissions. In *Proceedings of the 26th ACM international conference on Multimedia*. 1616–1624.

[2] Divyashri Bhat, Amr Rizk, and Michael Zink. 2017. Not so QUIC: A performance study of DASH over QUIC. In *Proceedings of the 27th workshop on network and operating systems support for digital audio and video*. 13–18.

[3] Kyunghyun Cho, Bart Van Merriënboer, Çaglar Gülçehre, Fethi Bougares, Holger Schwenk, and Yoshua Bengio. 2014. Learning phrase representations using RNN encoder-decoder for statistical machine translation. CoRR abs/1406.1078 (2014).

[4] Federal Communications Commission et al. 2016. Raw data-measuring broadband america. *Retrieved June* 19 (2016), 2018.

[5] Alireza Erfanian, Farzad Tashtarian, and Mohammad Hossein Yaghmaee. 2018. On maximizing QoE in AVC-based HTTP adaptive streaming: An SDN approach. In *2018 IEEE/ACM 26th International Symposium on Quality of Service (IWQoS)*. IEEE, 1–10.

[6] DASH Industry Form. 2016. Reference Client 2.4. 0.

[7] Tianchi Huang, Chao Zhou, Rui-Xiao Zhang, Chenglei Wu, Xin Yao, and Lifeng Sun. 2020. Stick: A Harmonious Fusion of Buffer-based and Learning-based Approach for Adaptive Streaming. In *IEEE INFOCOM 2020-IEEE Conference on Computer Communications*. IEEE, 1967–1976.

[8] Te-Yuan Huang, Ramesh Johari, Nick McKeown, Matthew Trunnell, and Mark Watson. 2014. A buffer-based approach to rate adaptation: Evidence from a large video streaming service. In *Proceedings of the 2014 ACM conference on SIGCOMM*. 187–198.

[9] Junchen Jiang, Vyas Sekar, and Hui Zhang. 2012. Improving fairness, efficiency, and stability in http-based adaptive video streaming with festive. In *Proceedings of the 8th international conference on Emerging networking experiments and technologies*. 97–108.

[10] Adam Langley, Alistair Riddoch, Alyssa Wilk, Antonio Vicente, Charles Krasic, Dan Zhang, Fan Yang, Fedor Kouranov, Ian Swett, Janardhan Iyengar, et al. 2017. The quic transport protocol: Design and internet-scale deployment. In *Proceedings of the conference of the ACM special interest group on data communication*. 183–196.

[11] Hongzi Mao, Ravi Netravali, and Mohammad Alizadeh. 2017. Neural adaptive video streaming with pensieve. In *Proceedings of the Conference of the ACM Special Interest Group on Data Communication*. 197–210.

[12] Lifan Mei, Runchen Hu, Houwei Cao, Yong Liu, Zifan Han, Feng Li, and Jin Li. 2020. Realtime mobile bandwidth prediction using LSTM neural network and Bayesian fusion. *Computer Networks* 182 (2020), 107515.

[13] Ravi Netravali, Anirudh Sivaraman, Somak Das, Ameesh Goyal, Keith Winstein, James Mickens, and Hari Balakrishnan. 2015. Mahimahi: Accurate Record-and-Replay for HTTP. In *2015 USENIX Annual Technical Conference (USENIX ATC 15)*. USENIX Association, Santa Clara, CA, 417–429. https://www.usenix.org/conference/atc15/technical-session/presentation/netravali

[14] Cisco Visual Networking. 2016. Cisco global cloud index: Forecast and methodology, 2016–2021. *White paper. Cisco Public, San Jose* 1 (2016).

[15] Ashkan Nikravesh, Yihua Guo, Xiao Zhu, Feng Qian, and Z Morley Mao. 2019. MP-H2: a Client-only Multipath Solution for HTTP/2. In *The 25th Annual International Conference on Mobile Computing and Networking*. 1–16.

[16] Mirko Palmer, Thorben Krüger, Balakrishnan Chandrasekaran, and Anja Feldmann. 2018. The quic fix for optimal video streaming. In *Proceedings of the Workshop on the Evolution, Performance, and Interoperability of QUIC*. 43–49.

[17] Pablo Gil Pereira, Andreas Schmidt, and Thorsten Herfet. 2018. Cross-layer effects on training neural algorithms for video streaming. In *Proceedings of the 28th ACM SIGMM Workshop on Network and Operating Systems Support for Digital Audio and Video*. 43–48.

[18] Darijo Raca, Jason J Quinlan, Ahmed H Zahran, and Cormac J Sreenan. 2018. Beyond throughput: a 4G LTE dataset with channel and context metrics. In *Proceedings of the 9th ACM Multimedia Systems Conference*. 460–465.

[19] Haakon Riiser, Paul Vigmostad, Carsten Griwodz, and Pål Halvorsen. 2013. Commute path bandwidth traces from 3G networks: analysis and applications. In *Proceedings of the 4th ACM Multimedia Systems Conference*. 114–118.

[20] Lihua Ruan, Maluge Pubuduni Imali Dias, and Elaine Wong. 2019. Machine learning-based bandwidth prediction for low-latency H2M applications. *IEEE Internet of Things Journal* 6, 2 (2019), 3743–3752.

[21] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. 2017. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347* (2017).

[22] Satadal Sengupta, Niloy Ganguly, Sandip Chakraborty, and Pradipta De. 2018. HotDASH: Hotspot aware adaptive video streaming using deep reinforcement learning. In *2018 IEEE 26th International Conference on Network Protocols (ICNP)*. IEEE, 165–175.

[23] Ashkan Sobhani, Abdulsalam Yassine, and Shervin Shirmohammadi. 2017. A video bitrate adaptation and prediction mechanism for HTTP adaptive streaming. *ACM Transactions on Multimedia Computing, Communications, and Applications (TOMM)* 13, 2 (2017), 1–25.

[24] Huyen TT Tran, Duc V Nguyen, Duong D Nguyen, Nam Pham Ngoc, and Truong Cong Thang. 2019. An LSTM-based approach for overall quality prediction in HTTP adaptive streaming. In *IEEE INFOCOM 2019-IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS)*. IEEE, 702–707.

[25] Jeroen Van Der Hooft, Stefano Petrangeli, Tim Wauters, Rafael Huysegems, Patrice Rondao Alface, Tom Bostoen, and Filip De Turck. 2016. HTTP/2-based adaptive streaming of HEVC video over 4G/LTE networks. *IEEE Communications Letters* 20, 11 (2016), 2177–2180.

[26] Bo Wei, Kenji Kanai, Wataru Kawakami, and Jiro Katto. 2018. HOAH: A hybrid TCP throughput prediction with autoregressive model and hidden Markov model for mobile networks. *IEICE Transactions on Communications* (2018), 2017CQP0007.

[27] Xiaoqi Yin, Abhishek Jindal, Vyas Sekar, and Bruno Sinopoli. 2015. A control-theoretic approach for dynamic adaptive video streaming over HTTP. In *Proceedings of the 2015 ACM Conference on Special Interest Group on Data Communication*. 325–338.

[28] Chaoqun Yue, Ruofan Jin, Kyoungwon Suh, Yanyuan Qin, Bing Wang, and Wei Wei. 2017. LinkForecast: cellular link bandwidth prediction in LTE networks. *IEEE Transactions on Mobile Computing* 17, 7 (2017), 1582–1594.

[29] Xiaofei Zhang, Thi Mai Trang Nguyen, and Guy Pujolle. 2012. Kalman filter based bandwidth estimation and predictive flow distribution for concurrent multipath transfer in wireless networks. In *2012 3rd IEEE International Conference on Network Infrastructure and Digital Content*. IEEE, 305–309.